

Where Prices Come From: The Interaction of Supply and Demand

ECON 101 - Macroeconomics

Muhebullah Karimzada

Fall 2026

- **3.1 The Demand Side of the Market**
- 3.2 The Supply Side of the Market
- 3.3 Market Equilibrium: Putting Buyers and Sellers Together
- 3.4 The Effect of Demand and Supply Shifts on Equilibrium

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3.1 Learning Objective

Goal

Discuss the variables that influence demand.

- We begin by investigating how **buyers** behave.
- **Market demand:** the demand by all the consumers of a given good or service.
- Key tools:
 - Demand schedule
 - Demand curve

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Figure 3.1 – A Demand Schedule and Demand Curve (1 of 3)

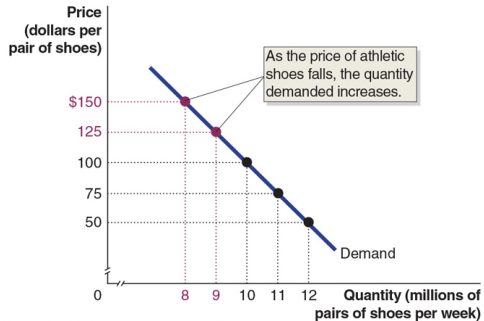
- **Demand schedule:** a table that shows the relationship between the **price** of a product and the **quantity of the product demanded**.
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Demand Schedule	
Price (dollars per pair of shoes)	Quantity (millions of pairs of shoes per week)
\$150	8
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100	10
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Figure 3.1 – Demand Curve and Ceteris Paribus (2 of 3)

- When drawing the demand curve, we assume **ceteris paribus**.
- **Ceteris paribus** (“all else equal”) condition:
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- **Quantity demanded:** the amount of a good or service that a consumer is willing and able to purchase at a given price.
- **Law of demand:**
 - Holding everything else constant, when the price of a product falls, the quantity demanded of the product will **increase**.
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What Explains the Law of Demand?

- When the price of a good falls, two key effects occur:
 - **Substitution effect:** consumers substitute toward the good whose price has fallen.
 - **Income effect:** consumers have more purchasing power, which is like an increase in income.

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Substitution and Income Effects – Definitions

- **Substitution effect:**

- The change in the quantity demanded of a good that results from a change in price, making the good more or less expensive **relative to other goods**, holding constant the effect of the price change on consumer purchasing power.

- **Income effect:**

- The change in the quantity demanded of a good that results from the effect of a change in the good's price on consumers' **purchasing power**, holding all other factors constant.

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Everyday Example: Movie Tickets

- Suppose movie ticket prices fall from \$15 to \$10.
- **Substitution effect:**
 - Movies are cheaper relative to other entertainment (e.g., bowling, dining out), so you substitute toward movies.
- **Income effect:**
 - You effectively have more real income; your budget goes further, so you can afford more movie nights per month.

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Figure 3.2 – Shifting the Demand Curve (1 of 2)

- A change in something **other than price** that affects demand causes the **entire demand curve to shift**.
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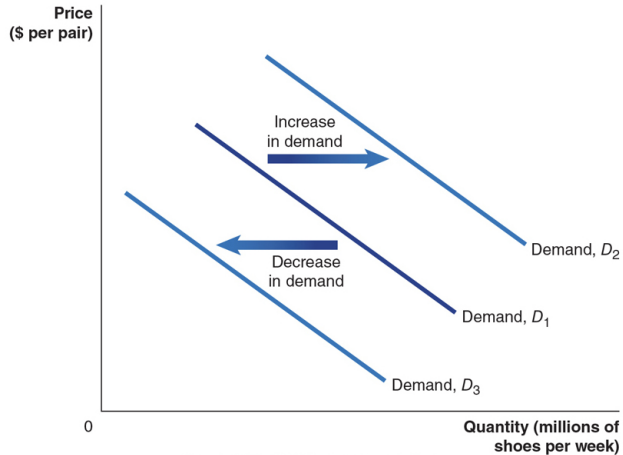
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Variables That Shift Market Demand

- Income of consumers
- Prices of related goods
- Tastes
- Population and demographics
- Expectations

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Changes in Income of Consumers

- **Normal goods:** goods for which demand increases as income rises and decreases as income falls.
 - Examples: new clothes, restaurant meals, vacations.
- **Inferior goods:** goods for which demand increases as income falls and decreases as income rises.
 - Examples: second-hand clothes, instant noodles.

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Effects of Changes in Income

- An increase in income will:
 - Increase the demand for new clothes (normal good), *ceteris paribus*.
 - Decrease the demand for second-hand clothes (inferior good).

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Apply the Concept: Lobster as Inferior vs. Normal Good

- In some parts of the Maritimes, lobster is an **inferior good**: as incomes rise, people eat *less* lobster.
- In large cities, lobster is a **normal good**: as incomes rise, people eat *more* lobster.
- Key lesson:
 - Whether a good is normal or inferior depends on **consumer behaviour** and context, not its intrinsic characteristics.

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Changes in the Price of Related Goods

- **Substitutes:** goods and services that can be used for the same purpose.
 - Examples: Big Mac and Whopper; Ford F-150 and Dodge Ram; jeans and khakis.
- **Complements:** goods and services that are used together.
 - Examples: Big Mac and McDonald's fries; hot dogs and hot dog buns.

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Effects of Changes in the Price of Related Goods

- An increase in the price of a Big Mac:
 - **Increases** the demand for Whoppers (substitute).
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Changes in Population and Demographics

- **Demographics:** characteristics of a population (age, race, gender, etc.).
- Increases in the **number of people** buying something will increase the amount demanded.
- Example:
 - An increase in the elderly population increases the demand for medical care.

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Changes in Expectations About Future Prices

- Consumers decide not only **which** products to buy but also **when** to buy them.
- Future products are **substitutes** for current products.
- An expected **increase** in price tomorrow $\Rightarrow$ increases demand today.
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Figure 3.3 – Change in Demand vs. Change in Quantity Demanded

- A change in the **price of the product itself** causes a **movement along** the demand curve:
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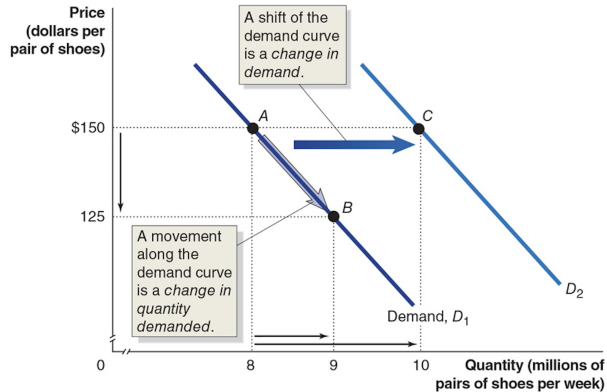
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- Now we examine **market supply**: decisions of firms about how much of a product to provide at various prices.

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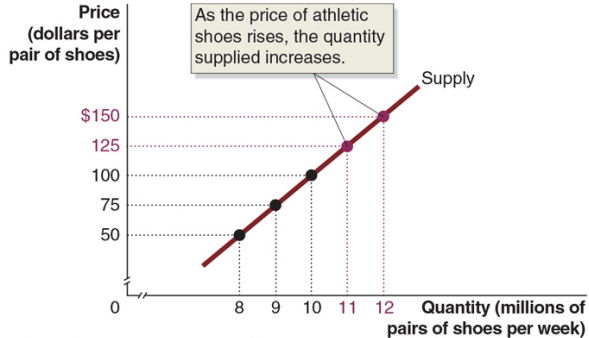
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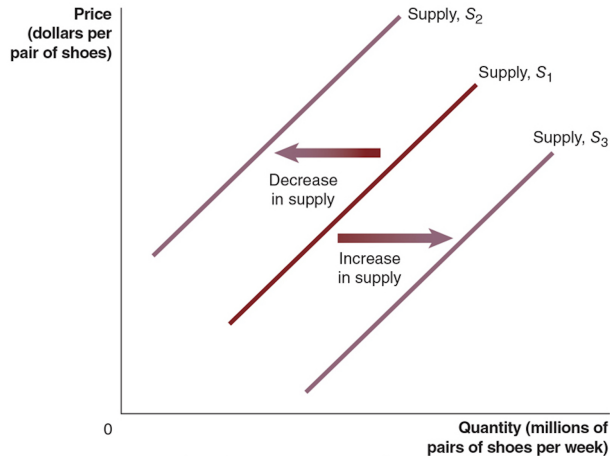
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- Technological change
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Changes in Prices of Inputs

- **Inputs** are anything used in the production of a good or service.
 - For athletic shoes: rubber, plastic, and labour.
- An **increase** in the price of an input decreases the profitability of selling the good, causing a **decrease in supply**.
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- A **decrease** in the price of an input increases profitability, causing an **increase in supply**.

Technological Change

- A firm may experience a positive or negative change in its ability to produce a given level of output with a given quantity of inputs.
- This is called **technological change**.
- Examples:
 - A new, more efficient way of producing shoes $\Rightarrow$ **increases** supply.
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Prices of Substitutes and Complements in Production

- Many firms can produce and sell alternative products: **substitutes in production**.
- Example:
 - An Ontario farmer can plant either corn or soybeans.
 - If the price of soybeans rises, the farmer will plant (supply) less corn.
- Sometimes two products are necessarily produced together: **complements in production**.
- Example:
 - Cattle provide both beef and leather.
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Figure 3.6 – Change in Supply vs. Change in Quantity Supplied

- A change in the **price of the product itself** causes a **movement along** the supply curve:
 - This is a **change in quantity supplied**.
- Any other change affecting supply shifts the entire supply curve:
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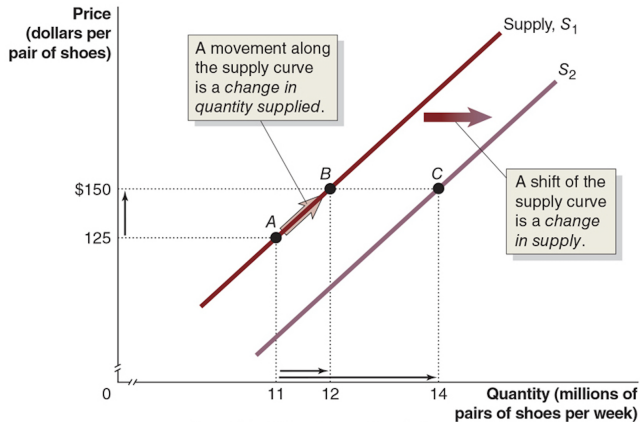
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3.3 Learning Objective

Goal

Use a graph to illustrate market equilibrium.

- **Market equilibrium:** quantity demanded equals quantity supplied.
- In markets with many buyers and sellers (perfectly competitive), we call this a **competitive market equilibrium**.

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- Consider the market for athletic shoes.
- At a price of \$100:
 - Consumers want to buy 10 million pairs of shoes per week.
 - Producers want to sell 10 million pairs of shoes per week.
- The **equilibrium price** is \$100, and the **equilibrium quantity** is 10 million pairs per week.

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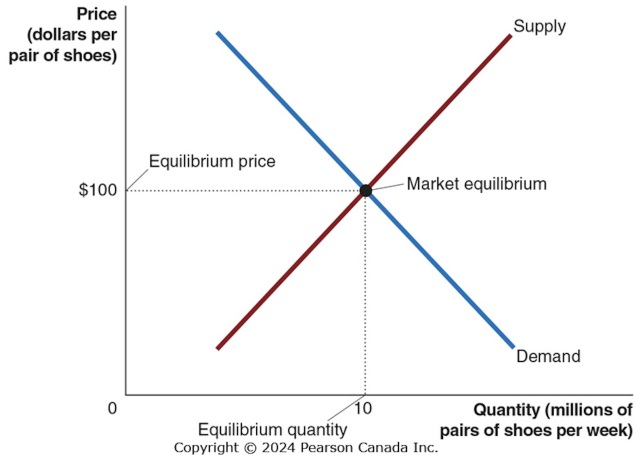


Figure 3.8 – Surpluses and Shortages (1 of 2)

- What if the price were \$125 instead?
- At \$125:
 - Consumers want to buy 9 million pairs of shoes.
 - Producers want to sell 11 million pairs of shoes.
- This gives a **surplus** of 2 million pairs: quantity supplied exceeds quantity demanded.
- Prediction: sellers will compete among themselves, driving the price **down** toward equilibrium.

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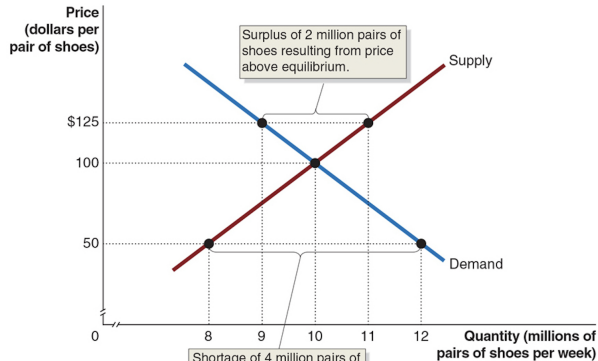
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Figure 3.8 – Surplus



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Figure 3.8 – Surpluses and Shortages (2 of 2)

- What if the price were \$50?
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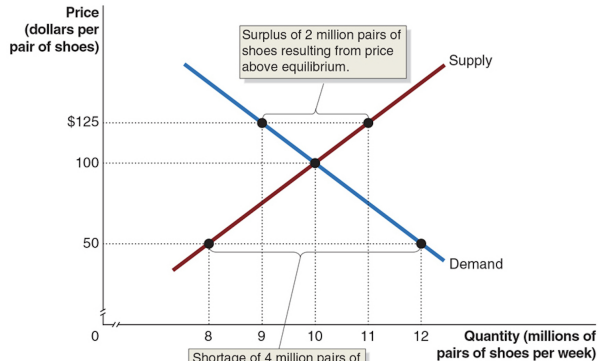
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Demand and Supply Both Count

- Price is determined by the **interaction** of buyers and sellers.
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Use demand and supply graphs to predict changes in prices and quantities.

- We typically know the actual price and quantity, but do *not* observe the whole curves.
- The power of the model is in predicting **directional changes** in price and quantity when curves shift.

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Figure 3.9 – Effect of an Increase in Supply (1 of 2)

- Market for athletic shoes when a new shoemaker enters the market.
- At any given price, more shoes are supplied:
 - Supply shifts right from S_1 to S_2 .
- Result:
 - Equilibrium price falls.
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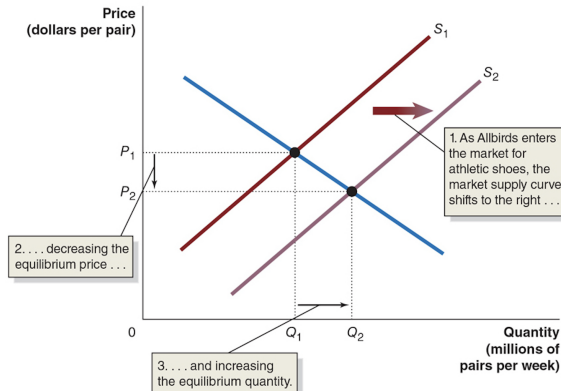
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Figure 3.9 – How Much Does Price Fall? (2 of 2)

- By how much will price fall?
- By how much will quantity rise?
- We **cannot** say without knowing more (the exact shapes and positions of demand and supply).
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Figure 3.10 – Effect of an Increase in Demand

- Suppose incomes increase.
- Athletic shoes are a **normal good**, so as income rises, demand shifts to the right.
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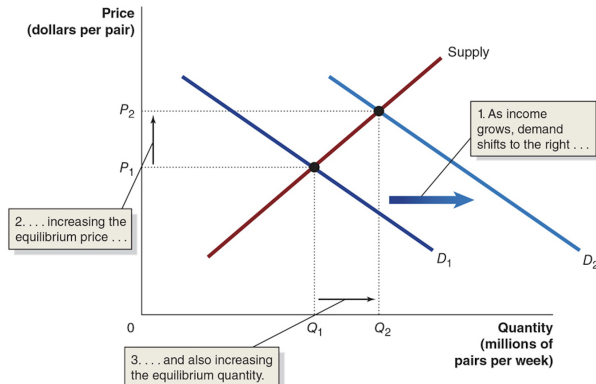
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Table 3.3 – How Shifts in Demand and Supply Affect Equilibrium P and Q

(1 of 2)

	Equilibrium Price P	Equilibrium Quantity Q
Increase in demand, supply unchanged	Rises	Rises
Decrease in demand, supply unchanged	Falls	Falls
Increase in supply, demand unchanged	Falls	Rises
Decrease in supply, demand unchanged	Rises	Falls

Table 3.3 – How Shifts in Demand and Supply Affect P and Q (2 of 2)

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Increase in demand, increase in supply	?	Rises
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- The highlighted row (in blue text) is the example we studied: **increase in supply**.
- “?” means the effect is **ambiguous** without more information on the relative size of shifts.

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Figure 3.11 – Shifts in Demand and Supply Over Time (1 of 3)

- Over time, it is likely that **both** demand and supply will change.
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 - New firms enter $\Rightarrow$ supply shifts right.
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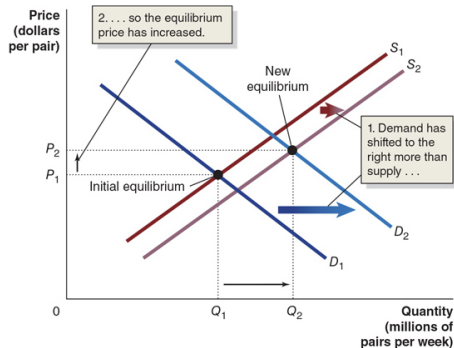
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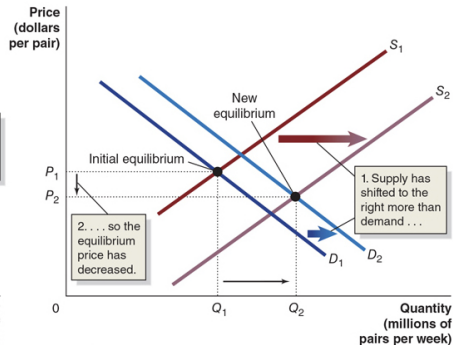
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Figure 3.11 – Both Demand and Supply Shift



(a) Demand shifting more than supply



(b) Supply shifting more than demand

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Figure 3.11 – Case 1: Demand Shifts More than Supply (2 of 3)

- When demand shifts more than supply:
 - Equilibrium quantity **rises**.
 - Equilibrium price also **rises**.
- This panel shows a larger rightward shift of demand than of supply.

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Figure 3.11 – Case 2: Supply Shifts More than Demand (3 of 3)

- When supply shifts more than demand:
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Shifts of Curves vs. Movements Along Curves

- Suppose an **increase in supply** occurs.
- We know:
 - Equilibrium quantity will increase.
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- It is tempting to say “the decrease in price causes an increase in demand”, but this is incorrect.
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 - Shifts when income, prices of related goods, tastes, population, or expectations change.
- Supply:
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